Adjacency matrices, walks, and connectivity: what the powers of A count and how A shows that a graph is connected

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Abstract

A finite simple undirected graph with N vertices can be given by its adjacency matrix A, a table of zeros and ones that records which pairs of vertices are joined by an edge. A *walk* is a route that follows edges from vertex to vertex and may revisit vertices and edges; a graph is *connected* if every vertex can be reached from every other. This note answers three questions about A, two taken from a lecture whiteboard and one from an assignment. First, the entry in row i and column j of the matrix power A^k counts the walks of length k from vertex n_i to vertex n_j; it counts walks, not paths. Second, the graph is connected if and only if every entry off the diagonal of $A + A^2 + \cdots + A^{N-1}$ is positive, which holds if and only if $(I + A)^{N-1}$ has no zero entry. Third, a path between n_i and n_j is a sequence of pairwise distinct vertices from n_i to n_j in which consecutive vertices are joined by an edge. The two tests and the walk-counting theorem are proved in full here; the textbook's definitions, the breadth-first-search comparison, and standard linear algebra are cited, not proved.

1 Introduction

1.1 The questions and where they came from

A graph is a set of points, called vertices, together with lines, called edges, that join some pairs of points. A computer cannot look at a drawing, so a graph is often given as a table of zeros and ones, the adjacency matrix A, with one row and one column for each vertex. This note is about what that table can tell us. It asks three questions.

(Q1) What does the entry in row i and column j of the matrix power $(A^k)_{ij}$ represent?

(Q2) How can we know from A whether the graph is connected, that is, whether every vertex can be reached from every other?

(Q3) What is the definition of a path between the vertices n_i and n_j?

Questions (Q1) and (Q2) come from a lecture whiteboard that was photographed twice from slightly different angles [5, 6]. The board holds a rule for the entries A_{ij} of an adjacency matrix, a drawing of three vertices, a 3×3 matrix with rows 0 1 0, 1 0 1, 0 1 0, a problem statement (a graph with N nodes is given through its adjacency matrix A), the two questions, and a definition: a graph is connected if for any n_i and n_j there exists a path between the nodes. Question (Q3) was added by the assignment, as it was relayed to the author of this note; the relay is a summary and not the assignment's own words [11].

The questions matter because they connect a picture to arithmetic. If multiplying the table by itself counts routes between vertices, then a routine calculation can answer questions that one would otherwise answer by looking at a drawing, such as whether every vertex can be reached from every other.

1.2 Conventions and assumptions

Where the board, the assignment as relayed, and the course textbook (Petrosyan, Mkrtchyan, and Kamalyan, *Graph Theory*, in Armenian) differ or leave a point open, this list fixes what the note uses. Later sections cite it by item number. The textbook is cited by printed page number; the page of the PDF file is one higher [10].

1. **Setting.** The graph is finite, simple (no edge joins a vertex to itself, and no two edges have the same two ends), and undirected, and it has $N \geq 1$ vertices. This is how the assignment was relayed [11]; it agrees with the board matrix, which is symmetric and has zeros on its diagonal [5].

2. **Names.** We write $G = (V, E)$ with V the vertices and E the edges, call the vertices $n_1, \ldots, n_N$ as the board does, and write $n_i n_j$ for the edge between n_i and n_j. The board uses other letters (its V is the set of edges, and it writes e_{ij} for an edge), and the textbook writes v_i for vertices and never e_{ij} [10, p. 9]. This is a difference of notation only.

3. **The exponent.** In question (Q1) the board has a small mark above A that could not be resolved in the photos as a letter k or n [5]. It is read as a power k, because the assignment, as relayed, asks about A^k [11]. The theorem in section 4 holds for every k.

4. **The matrix defines the example.** The board drawing and the board matrix do not show the same graph (section 6.2). Because the problem statement says that the graph is given through its adjacency matrix, the *matrix* is the example. Its graph has the two edges $n_1 n_2$ and $n_2 n_3$ and is called the *board graph*.

5. **The word "path".** In the course, "path" means a walk with no repeated vertex (a walk is a route along edges, which may revisit vertices and edges; definition 3.1). This is the textbook's Definition 1.2.13, the *simple path* [10]; that the course uses it is part of the relayed assignment [11]. The textbook's own word "path" means something else, which this note calls a *trail*; table 2 compares the words.

6. **Length 0.** A single vertex counts as a walk of length 0. This is chosen so that the identity matrix $A^0 = I$ counts walks too: from each vertex to itself there is one walk of length 0, and between different vertices there is none. The textbook's definitions do not say this in words, but its usage assumes it [10, pp. 23, 49], and Grinberg's definition of a walk allows it [3, Definition 2.9.1(a), p. 36]. It is this note's own convention.

7. **Who proves what.** As far as a search of its text showed, the textbook contains neither the walk-counting theorem nor a matrix test for connectivity (details in appendix A) [10]. This note proves both, and names in the first sentence after each statement the sources that also state it.

1.3 The story

One story is told throughout, and every plain-words paragraph uses only its words. A pond has some *islands*. Some pairs of islands have a *bridge* between them. A frog sits on an island and makes *hops*: each hop crosses one bridge to a *next-door* island. The frog may hop back, and may use the same bridge again. A square *table* has one row and one column for each island. A cell holds a one if the row island and the column island are next door to each other. Otherwise it holds a zero. Table 1 gives the dictionary. Each story word stands for one technical term. In the formal text, the plain-words paragraph comes first and uses only story words; the definition that follows gives the technical term.

1.4 The answers in brief

- (Q1) The entry in row i and column j of A^k is the number of walks of length k from n_i to n_j (theorem 4.1). It counts walks, not paths (example 4.5), and it is not simply 1 when n_j can be reached from n_i (non-example 4.3).

Story word	Technical term
island	vertex
bridge	edge
next door	adjacent
the number of bridges at an island	degree
table, cell	adjacency matrix, entry
trip, with a number of hops	walk, with its length
round trip	closed walk
fresh-bridge trip	trail
fresh-island trip	path
ring trip	cycle
fewest hops between two islands	distance
in one piece	connected
group of islands	connected component

Table 1: The story dictionary. One story word stands for one technical term.

- (Q2) G is connected if and only if every entry off the diagonal of $B = A + A^2 + \cdots + A^{N-1}$ is positive, which holds if and only if $(I + A)^{N-1}$ has no zero entry (theorems 5.12 and 5.13). A single power is not a test (example 5.16).
- (Q3) A path between n_i and n_j is a sequence $n_i = u_0, u_1, \ldots, u_k = n_j$ of pairwise distinct vertices in which consecutive vertices are adjacent (definition 3.17).

1.5 Organisation and reading guide

Section 2 fixes the graph, the adjacency matrix, and matrix powers. Section 3 introduces walks, trails, paths, and cycles in the textbook's order and answers (Q3). Section 4 answers (Q1) and section 5 answers (Q2), with worked examples, one connected and one not. Section 6 collects limits and the comparison with other sources, and section 7 states the answers. The appendix lists the sources, the pages checked, and how to rerun the scripts.

A reader who wants only the answers can read section 2 up to definition 2.18, then definitions 3.1, 3.13 and 3.17, lemma 3.30, and corollary 3.32 in section 3; theorem 4.1 in section 4; and definition 5.1, lemma 5.11, theorem 5.12, and the examples in section 5. Trails, cycles, distance, components, the power test, and section 6 are not needed for (Q1) to (Q3).

2 Graphs and the adjacency matrix

Notation. Throughout, $G = (V, E)$ is a finite simple undirected graph (Convention 1) with vertex set $V = \{n_1, \ldots, n_N\}$, $N \geq 1$, and edge set E. The letters u, v, w also stand for vertices; uv is the edge joining u and v, and $|E|$ is the number of edges. The indices i, j, r run over $1, \ldots, N$. The letter k is a non-negative integer used for the length of a walk and for the exponent of a matrix power. A is the adjacency matrix, I the $N \times N$ identity matrix, and $\deg(n_i)$ the degree of n_i. The textbook writes the same objects with v_i and $d_G(v)$ [10, pp. 9, 16]; this note writes deg for degree so that the letter d is free for distance (definition 3.25).

2.1 Graphs, neighbours, and the adjacency matrix

In plain words. A pond has some islands. Some pairs of islands have a bridge between them. A bridge joins two different islands. No two islands have more than one bridge. A bridge can be crossed in both directions.

Definition 2.1 (Graph; textbook Definition 1.1.1)**.** A *graph* G is an ordered pair (V, E), where V is a set whose elements are the *vertices* and E is a set whose elements are the *edges* [10, Definition 1.1.1, p. 9]. In this note V is finite, and each edge is a set $uv = \{u, v\}$ of two different vertices (Convention 1).

Remark 2.2. Because an edge is a set of two different vertices, there is no edge from a vertex to itself, and uv and vu are the same edge, so there is no direction. Because E is a set, two vertices are joined by at most one edge.

Example 2.3. The board graph has $V = \{n_1, n_2, n_3\}$ and $E = \{n_1n_2, n_2n_3\}$ (fig. 1, left).

Non-example 2.4. Giving n_1 an edge to itself, or giving n_1 and n_2 two separate edges, does not produce a graph in this note's sense.

In plain words. Two islands are next door when one bridge joins them. If a frog can hop between two islands in one hop, they are next door. An island is never next door to itself.

Definition 2.5 (Adjacent; textbook Definition 1.1.4)**.** The vertices u and v are *adjacent* if $uv \in E$ [10, Definition 1.1.4, p. 11].

Remark 2.6. Adjacency is symmetric, and in a simple graph no vertex is adjacent to itself.

Example 2.7. In the board graph, n_1 and n_2 are adjacent.

Non-example 2.8. In the board graph, n_1 and n_3 are not adjacent, and neither is n_2 with itself.

In plain words. Draw a square table. Give each island one row and one column. Put a one in a cell when the row island and the column island are next door. Put a zero in every other cell. The table lists all the bridges.

Definition 2.9 (Adjacency matrix; textbook, after Definition 1.1.6)**.** The *adjacency matrix* of G is the $N \times N$ matrix $A = (A_{ij})$ with

$$A_{ij} = \begin{cases} 1, & n_i \text{ and } n_j \text{ are adjacent,} \\ 0, & \text{otherwise.} \end{cases} \tag{1}$$

The textbook gives this matrix no number of its own; it states the rule and notes $a_{ii} = 0$ and $a_{ij} = a_{ji}$ on printed p. 11 [10]. The board writes the same rule as $A_{ij} = 1$ if $e_{ij} \in V$ and 0 if $e_{ij} \notin V$, with e_{ij} the edge joining n_i and n_j and V the board's name for the edge set [5, 6] (Convention 2).

Remark 2.10. Every entry is 0 or 1, never 2 or a weight, which is correct because a simple graph has at most one edge between two vertices. A 1 on the diagonal would be a loop, which a simple graph does not have.

Example 2.11. The board matrix has rows 0 1 0, 1 0 1, 0 1 0 [5, 6]:

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}.$$

It is the adjacency matrix of the board graph, whose only edges are n_1n_2 and n_2n_3 (Convention 4).

Non-example 2.12. The matrix with rows 0 1 0, 0 0 1, 0 1 0 is not the adjacency matrix of a simple undirected graph, because it is not symmetric. A matrix with a 1 in the top left corner is not one either, because that 1 would be a loop.

In plain words. Count the bridges that touch one island. That count is the number of bridges at the island.

Definition 2.13 (Degree; textbook Definition 1.2.1)**.** The *degree* $\deg(n_i)$ of the vertex n_i is the number of edges that have n_i as an end [10, Definition 1.2.1, p. 16].

Remark 2.14. The textbook defines degree after the adjacency matrix, so this note uses it only after definition 2.9.

Example 2.15. In the board graph the degrees of n_1, n_2, n_3 are $1, 2, 1$.

Non-example 2.16. The number of edges of the board graph, which is 2, is not the degree of n_1, which is 1.

In plain words. Flip the table over its slanted line of cells from top left to bottom right. Nothing changes. Cells on that line stay zero, because no island is next door to itself. Add up one row. You get the number of bridges at that island. Add up the whole table. You count every bridge twice.

Proposition 2.17 (Basic properties of A)**.** *Let A be the adjacency matrix of G. Then*

1. $A_{ij} = A_{ji}$ *for all i, j (symmetry);*
2. $A_{ii} = 0$ *for all i (zero diagonal);*
3. $\sum_j A_{ij} = \deg(n_i)$ *for every i (row sums are degrees);*
4. $\sum_{i,j} A_{ij} = 2|E|$.

Items 1 and 2 are stated in the textbook [10, p. 11] and in the University of Galway notes [7, Notes 1 and 2, p. 3]; the proof below is the note's own.

Proof. (1) n_i and n_j are adjacent exactly when n_j and n_i are, since $uv = vu$. (2) A simple graph has no edge from n_i to itself, so n_i is not adjacent to itself. (3) Row i has a 1 in column j exactly when $n_i n_j \in E$, so the row sum counts the edges at n_i, which is $\deg(n_i)$. (4) By (3) the sum of all entries is the sum of all degrees. Every edge has two ends, so it is counted once at each end when the degrees are added; the sum is therefore $2|E|$, which is the textbook's Theorem 1.2.1 (Euler) [10, Theorem 1.2.1, p. 16]. Since A is symmetric, column sums are the degrees too. □

2.2 Matrix products and powers

In plain words. To make a new table from two tables, take one row of the first and one column of the second. Multiply their cells in pairs. Add the results. That is one cell of the new table. Do this for every row and column. Do it to the island table with itself, and again, and again.

Definition 2.18 (Matrix product and power)**.** For $N \times N$ matrices X and Y the product XY has entries

$$(XY)_{ij} = \sum_{r=1}^{N} X_{ir} Y_{rj}. \tag{2}$$

The *powers* of A are $A^0 = I$ (ones on the diagonal, zeros elsewhere) and $A^{k+1} = A^k A$ for $k \geq 0$. This is the standard definition from linear algebra. In words, the entry in row i and column j of XY is obtained by running along row i of X and down column j of Y, multiplying the entries in pairs, and adding the results. The *transpose* X^T of X is the matrix with $(X^T)_{ij} = X_{ji}$, that is, X flipped over its diagonal; the standard rule $(XY)^T = Y^T X^T$ is used in section 4. By proposition 2.17, $A^T = A$.

Remark 2.19. A power is not the entrywise power. Every entry of A^k is a non-negative integer, because A has entries 0 and 1 and sums and products of non-negative integers are non-negative integers; this is used in section 5.

Example 2.20. For the board matrix, row 1 of A is $(0,1,0)$ and column 1 of A is $(0,1,0)$, so $(A^2)_{11} = 0 \cdot 0 + 1 \cdot 1 + 0 \cdot 0 = 1$. Row 2 is $(1,0,1)$ and column 2 is $(1,0,1)$, so $(A^2)_{22} = 1 \cdot 1 + 0 \cdot 0 + 1 \cdot 1 = 2$.

Non-example 2.21. Squaring every entry of A separately gives A back, because the entries are 0 or 1; it is not A^2, whose entry in row 2 and column 2 is 2, not 0.

3 Walks, trails, paths, and cycles

This section answers question (Q3), the definition of a path between n_i and n_j, in the order in which the textbook builds the concepts [10, pp. 21–23]: walk, closed walk, trail, path, cycle, distance. Each concept is a small change to the one before it. The examples use the graph H on five vertices with edges n_1n_2, n_2n_3, n_1n_3, n_3n_4, n_4n_5 (fig. 1, middle), which is this note's own example, not the textbook's; the classification of every sequence below was checked by script (appendix A).

Only walks and paths (definitions 3.1, 3.13 and 3.17) and the lemmas on them are needed for (Q1) to (Q3). Trails are included because the textbook uses the word "path" for them and its definition of connected rests on that word; cycles and distance are included to keep the textbook's order, and distance is used in corollary 4.7.

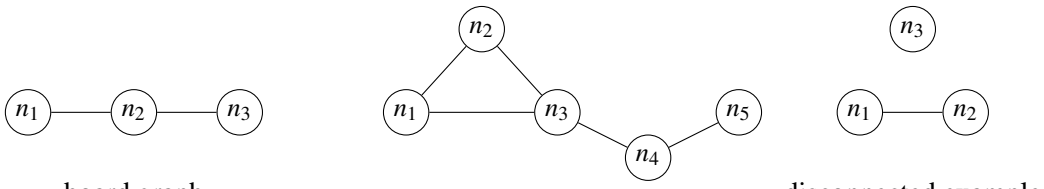


Figure 1: The graphs used in the examples. Left: the board graph, with edges n_1n_2 and n_2n_3 and the adjacency matrix of the board (Convention 4). Middle: the graph H of this section, with edges n_1n_2, n_2n_3, n_1n_3, n_3n_4, n_4n_5. Right: one edge and an isolated vertex, the disconnected example of section 5.

Notation. A (u,v)-*walk* is a walk from u to v. In examples, a sequence of vertices such as $n_1n_2n_3$ is written without commas.

3.1 The definitions

In plain words. A frog sits on an island. It hops over a bridge to a next-door island, then hops again. The list of islands it visits is a trip. A trip may land on an island again. It may use a bridge again. The number of hops is how long the trip is. A frog that does not hop at all makes a trip with no hops.

Definition 3.1 (Walk; textbook Definition 1.2.10)**.** A *walk from* u_0 *to* u_k is a sequence of vertices $u_0u_1\cdots u_k$ in which each vertex is adjacent to the one before it; its *length* is k, the number of edges used. Formally, the textbook defines it as a sequence $u_0, e_1, u_1, \ldots, e_k, u_k$ of vertices and edges with $e_t = u_{t-1}u_t$ for $1 \le t \le k$ [10, Definition 1.2.10, p. 21]; in a simple graph the edges are fixed by the vertices, so the vertex sequence says the same. This note allows $k = 0$ (Convention 6), which the textbook's definition does not state in words.

Remark 3.2. Vertices and edges may repeat, and the length counts each edge as often as it is used [10, p. 22]. A single vertex u is a walk of length 0 from u to u (Convention 6).

Example 3.3. In H, the sequence $n_1n_2n_3n_4n_3n_1$ is a walk of length 5.

Non-example 3.4. In H, the sequence n_1n_4 is not a walk, because n_1n_4 is not an edge.

In plain words. A round trip ends on the island where it began.

Definition 3.5 (Closed walk; textbook Definition 1.2.11)**.** A walk $u_0\cdots u_k$ is *closed* if $u_0 = u_k$ [10, Definition 1.2.11, p. 22].

Remark 3.6. A walk of length 0 is closed, and so is a walk that goes along one edge and straight back.

Example 3.7. In H, the walk $n_1n_2n_1$ is closed.

Non-example 3.8. In H, the walk $n_1n_2n_3$ is not closed.

In plain words. A fresh-bridge trip never uses the same bridge twice. It may land on an island again.

Definition 3.9 (Trail; textbook Definition 1.2.12, where it is called "path")**.** A walk is a *trail* if its edges $u_0u_1, \ldots, u_{k-1}u_k$ are pairwise distinct. Its length is its number of edges [10, Definition 1.2.12, p. 22].

Remark 3.10. The textbook's word for a trail is "path" (Convention 5), which is not what this note means by "path". In a trail vertices may repeat but edges may not. The English word "trail" for this notion is used by Grinberg [3, Definition 3.4.5, p. 101]; it is this note's choice for the textbook's concept, because the course's word for it was not reported.

Example 3.11. In H, the walk $n_3n_1n_2n_3n_4$ is a trail: its edges n_3n_1, n_1n_2, n_2n_3, n_3n_4 are distinct, although n_3 occurs twice.

Non-example 3.12. In H, the walk $n_1n_2n_1n_3$ is not a trail, because it uses the edge n_1n_2 twice.

In plain words. A fresh-island trip never lands on the same island twice.

Definition 3.13 (Path; textbook Definition 1.2.13, the "simple path")**.** A walk is a *path* if its vertices $u_0, \ldots, u_k$ are pairwise distinct. This is the textbook's simple path, defined there as a textbook "path" (a trail) with distinct vertices [10, Definition 1.2.13, p. 22]; it is the course convention (Convention 5).

Remark 3.14. Every path is a trail, because distinct vertices force distinct edges (lemma 3.29); every trail is a walk; the converses fail. This is why the definition here needs only "no repeated vertex", whereas the textbook states it as a trail with distinct vertices.

Example 3.15. In H, the walk $n_1n_3n_4n_5$ is a path of length 3, and so is $n_1n_2n_3n_4n_5$, of length 4.

Non-example 3.16. In H, the trail $n_3n_1n_2n_3n_4$ is not a path, because n_3 occurs twice. The walk $n_1n_2n_1$ is not a path either.

In plain words. Pick two islands. Sometimes the frog can get from the first to the second. If it lands on no island twice, that is a fresh-island trip between them. A frog that stays put makes such a trip from an island to itself.

Definition 3.17 (Path between n_i and n_j; the answer to (Q3))**.** Let $n_i, n_j \in V$. A *path between n_i and n_j* (also a path *from n_i to n_j*) is a sequence $u_0, u_1, \ldots, u_k$ with $k \geq 0$, $u_0 = n_i$, $u_k = n_j$, such that u_{t-1} and u_t are adjacent for $1 \leq t \leq k$ and $u_0, \ldots, u_k$ are pairwise distinct. Its length is k. This is definition 3.13 with the ends fixed, that is, the textbook's Definition 1.2.13 with ends n_i and n_j.

Remark 3.18. If $i \neq j$, such a path has at least one edge. If $i = j$, the only path is the single vertex (n_i) of length 0, because $u_0 = u_k$ with $k \geq 1$ would repeat a vertex. Two vertices may be joined by several paths of different lengths.

Example 3.19. In the board graph the only path from n_1 to n_3 is $n_1n_2n_3$, of length 2. In H there are two paths from n_1 to n_4: $n_1n_3n_4$ and $n_1n_2n_3n_4$.

Non-example 3.20. In the board graph, $n_1n_2n_1n_2n_3$ is a walk from n_1 to n_3 but not a path, because it repeats n_1. And $n_1n_2n_1$ is not a path from n_1 to n_1.

In plain words. Some round trips never use a bridge twice. If such a round trip also lands twice on no island except the first, we call it a ring trip. A ring trip needs at least three bridges.

Definition 3.21 (Closed trail and cycle; textbook Definitions 1.2.14 and 1.2.15)**.** A *closed trail* is a closed walk that is a trail. The textbook calls it a "closed path or cycle" [10, Definition 1.2.14, p. 22]. A *cycle* is a closed trail of positive length whose only repeated vertices are its first and last; this is the textbook's simple cycle [10, Definition 1.2.15, p. 22] and the English "cycle". The word "cycle" is used only in this sense.

Remark 3.22. In a simple graph a cycle has length at least 3. Length 1 would need an edge from a vertex to itself. Length 2 would be $w_0w_1w_0$, which uses the edge w_0w_1 twice and is not a trail. The textbook's C_n has $n \geq 3$ [10, p. 22]. A closed trail need not be a cycle: in the graph with edges n_1n_2, n_2n_3, n_1n_3, n_1n_4, n_4n_5, n_1n_5 (two triangles sharing n_1), the sequence $n_1n_2n_3n_1n_4n_5n_1$ is a closed trail of length 6 that repeats n_1 in the middle.

Example 3.23. In H, the walk $n_1n_2n_3n_1$ is a cycle of length 3.

Non-example 3.24. In H, the walk $n_1n_2n_1$ is a closed walk but not a closed trail. The board graph has no cycle at all.

In plain words. The fewest hops between two islands is how far apart they are. If the frog can never get from one to the other, we say they are infinitely far apart.

Definition 3.25 (Distance; textbook Definition 1.2.16)**.** The *distance* $d(u,v)$ is the length of a shortest (u,v)-trail if there is one, and ∞ otherwise [10, Definition 1.2.16, pp. 22–23]. (The textbook words this with its own "path", that is, trail.)

Remark 3.26. The textbook measures distance by shortest trails. Proposition 3.34 shows that shortest walks and shortest paths give the same number, so this note may say "shortest path" without changing the textbook's value.

Example 3.27. In the board graph, $d(n_1,n_3) = 2$ and $d(n_1,n_1) = 0$.

Non-example 3.28. Add an isolated vertex n_4 to the board graph. Then $d(n_1,n_4) = \infty$, not 0 or 3.

Concept	This note (course)	Textbook's word	English	Textbook number
walk, length $k \geq 0$	walk	walk	walk	1.2.10
walk with first vertex equal to last	closed walk	closed walk	closed walk	1.2.11
walk with no repeated edge	trail	"path"	trail	1.2.12
walk with no repeated vertex	path	"simple path"	path	1.2.13
closed trail	closed trail	"closed path or cycle"	closed trail	1.2.14
closed trail, only first and last vertex repeat	cycle	"simple cycle"	cycle	1.2.15
least length of such a walk	distance	distance	distance	1.2.16

Table 2: Three naming conventions for the same concepts. The textbook column is from [10, pp. 21–23]. The English column is the usage of Grinberg [3, Definitions 2.9.1(c), 2.10.1(b), 3.4.5] and of Bai, who defines a path as a walk where no vertex is repeated [1, p. 1]; it is the usage of those two sources, not a survey of English books. The word "trail" in the middle column is this note's choice (Convention 5).

3.2 Results about walks and paths

In plain words. A fresh-island trip never lands on an island twice. So it cannot use the same bridge twice either.

Lemma 3.29 (Distinct vertices give distinct edges)**.** *A walk $u_0 \cdots u_k$ whose vertices are pairwise distinct has pairwise distinct edges. Hence every path is a trail.*

The textbook's Definition 1.2.13 asks for both properties [10, p. 22]; the proof is the note's own.

Proof. Suppose two edges are equal, $e_s = e_t$ with positions $1 \le s < t \le k$, where $e_s = \{u_{s-1}, u_s\}$ and $e_t = \{u_{t-1}, u_t\}$. The vertex u_{s-1} is an end of e_s, hence an end of e_t, so $u_{s-1} = u_{t-1}$ or $u_{s-1} = u_t$. Because the vertices of the walk are pairwise distinct, two of them are equal only if they have the same position, so $s-1 = t-1$ or $s-1 = t$. Both are impossible, because $s < t$. So no two edges are equal. □

In plain words. Take any trip from one island to another. Suppose the frog lands on some island twice. Cut out all the hops between the two visits. What is left is still a trip between the same two islands. It has fewer hops. Keep cutting. In the end the frog lands on no island twice.

Lemma 3.30 (Every walk contains a path)**.** *Let $W = u_0 \cdots u_k$ be a walk from u to v (equal or not). Then there is a path from u to v of length at most k whose edges are edges of W.*

The textbook's Lemma 1.2.1 states this for $u \neq v$, without a bound on the length, and proves it by induction on the length of the walk [10, Lemma 1.2.1, p. 23]. Grinberg's Corollary 2.9.7 covers $u = v$ and gives the bound [3, p. 39]. The proof below, which allows $k = 0$, is the note's own.

Proof. By strong induction on k: to prove the statement for walks of length k, we may assume it for all shorter walks, of every length from 0 to $k-1$. (Ordinary induction would assume it only for length $k-1$; we need more, because the cut below removes an unknown number of edges.) *Base case* $k = 0$. Then $W = (u)$ has one vertex, which is not repeated, so W is itself a path of length 0. *Inductive case.* Let $k \ge 1$ and assume the statement for all walks shorter than k. If the vertices of W are pairwise distinct, W is a path of length k. Otherwise there are positions $s < t$ with $u_s = u_t$, the same vertex visited twice. Form $W' = u_0 \cdots u_s u_{t+1} \cdots u_k$, that is, delete the part of W between the two visits (if $t = k$, then $W' = u_0 \cdots u_s$). It is a walk, because the only new consecutive pair is u_s, u_{t+1} and $u_s = u_t$ is adjacent to u_{t+1}. It runs from u to v, its edges are edges of W, and its length is $k - (t-s) < k$. By the induction hypothesis W' contains a path from u to v of length at most $k - (t-s) \le k$. (The example after this proof shows the cut.) □

Example 3.31. In the board graph, cutting the walk $n_1 n_2 n_1 n_2 n_3$ at the repeated vertex n_1 leaves the path $n_1 n_2 n_3$; cutting the closed walk $n_1 n_2 n_3 n_2 n_1$ leaves the path n_1 of length 0.

In plain words. A fresh-island trip lands on each island at most once. So it cannot have as many hops as there are islands. With three islands, it has at most two hops.

Corollary 3.32 (Length bound)**.** *A path in a graph with N vertices has length at most $N-1$. Hence if some walk joins u to v, a path of length at most $N-1$ joins them.*

Proof. A path of length k has $k+1$ pairwise distinct vertices among the N vertices of the graph, so $k+1 \le N$ [3, Section 2.9.4, p. 40]. The second sentence follows with lemma 3.30: a walk of any length contains a path, and every path has length at most $N-1$. □

The bound is attained: the path graph on N vertices has a path of length $N-1$.

In plain words. Suppose some trip joins two islands. Then a fresh-bridge trip joins them, and a fresh-island trip joins them too. So it does not matter which kind of trip we ask for.

Proposition 3.33 (Walk, trail, and path exist together)**.** *For vertices u and v the following are equivalent: (a) a (u,v)-walk exists; (b) a (u,v)-trail exists; (c) a (u,v)-path exists.*

Grinberg's Proposition 2.9.10 states the equivalence of walks and paths only [3, p. 42]; the trail case is proved here.

Proof. (c) implies (b) by lemma 3.29. (b) implies (a), because a trail is a walk. (a) implies (c) by lemma 3.30. □

Consequently the textbook's definition of a connected graph, which asks for a textbook "path" (a trail), the board's definition, read with the course's path, and a definition by walks select the same graphs (section 5).
In plain words. The fewest hops over all trips between two islands is the same as the fewest hops over all fresh-bridge trips. It is also the same as the fewest hops over all fresh-island trips.

Proposition 3.34 (Distance)**.** *For vertices u and v let L_w, L_t, L_p be the least length of a walk, a trail, and a path from u to v (with ∞ if none exists). Then $L_w = L_t = L_p = d(u,v)$.*

The textbook lists three properties of distance without proof [10, p. 23]; the statement here and its proof are the note's own.

Proof. Every path is a trail and every trail is a walk (lemma 3.29), so $L_w \le L_t \le L_p$. If a walk W of least length L_w exists and repeated a vertex, the cut in the proof of lemma 3.30 would give a shorter walk, contradicting minimality; so W is a path and $L_p \le L_w$. The three numbers agree. The textbook's $d(u,v)$ is L_t by definition 3.25. If no walk exists, all three are ∞. □

Remark 3.35. The textbook's three properties of distance then follow from walks [10, p. 23]. First, $d(u,v) = 0$ exactly when $u = v$, because a walk of length 0 ends where it starts (this uses Convention 6). Second, $d(u,v) = d(v,u)$, because reversing a walk gives a walk of the same length. Third, $d(u,v) \le d(u,w) + d(w,v)$, because a walk from u to w followed by a walk from w to v is a walk from u to v whose length is the sum (trivially true when a distance is ∞).

4 Question 1: what the entries of A^k count

Question (Q1) asks what the entry in row i and column j of a power of A represents (Convention 3). The answer is a count of walks (definition 3.1); it is not a count of paths (definition 3.13).

4.1 The walk-counting theorem

In plain words. Look at the island table. A one means one hop is possible. To make two hops, the frog first goes to some middle island, and then on to the last island. For each middle island, multiply two cells and add the results. That is exactly how the table is multiplied by itself. So the table multiplied by itself counts two-hop trips. The same idea works for any number of hops.

Theorem 4.1 (Walk counting)**.** *Let G be a simple undirected graph with vertices $n_1, \dots, n_N$ and adjacency matrix A. For every integer $k \ge 0$ and all i, j, the entry $(A^k)_{ij}$ is the number of walks of length k from n_i to n_j.*

The University of Galway notes state the theorem for a positive integer k [7, Theorem 1.1.3, p. 4]; Bai states it for walks of length k [1, Theorem 1, p. 1], and Gotti for walks of length $\ell \in \mathbb{N}$ [2, Proposition 2, p. 1], and in the passages read neither says whether $k = 0$ is included. The Galway notes and Gotti prove it by induction on k. The case $k = 0$ is this note's convention (Convention 6). The textbook does not contain the theorem (Convention 7). The proof below is the note's own.

Proof. By induction on k.

Base case $k = 0$. $A^0 = I$. A walk of length 0 from n_i to n_j is a single vertex that is both the start and the end, so there is exactly one such walk if $i = j$, namely (n_i), and none if $i \ne j$. This is I_{ij}. (A reader who does not want length 0 can start at $k = 1$ instead: a walk of length 1 from n_i to n_j is the sequence $n_i n_j$ with n_i adjacent to n_j, so there is one if $A_{ij} = 1$ and none if $A_{ij} = 0$, which is A_{ij}. The inductive case below is unchanged.)

Induction hypothesis. Suppose that for some $k \ge 0$ and all pairs (i, r), the entry $(A^k)_{ir}$ is the number of walks of length k from n_i to n_r.

Inductive case. Fix i and j. Since $A^{k+1} = A^k A$, by eq. (2)

$$(A^{k+1})_{ij} = \sum_{r=1}^{N} (A^k)_{ir} A_{rj}. \tag{3}$$

Sort the walks of length $k+1$ from n_i to n_j by their last-but-one vertex. A walk $u_0 \cdots u_k u_{k+1}$ with $u_0 = n_i$ and $u_{k+1} = n_j$ has $u_k = n_r$ for exactly one r. Fix r.

- If n_r is adjacent to n_j (so $A_{rj} = 1$), the walks of length $k+1$ from n_i to n_j whose last-but-one vertex is n_r correspond one to one to the walks of length k from n_i to n_r: delete the final vertex n_j to go one way, and append n_j to go back, which gives a walk because n_r is adjacent to n_j. By the induction hypothesis there are $(A^k)_{ir}$ of them, and $(A^k)_{ir} = (A^k)_{ir} \cdot 1 = (A^k)_{ir} A_{rj}$.
- If n_r is not adjacent to n_j (so $A_{rj} = 0$), no walk can end with the move from n_r to n_j, so there are $0 = (A^k)_{ir} \cdot 0 = (A^k)_{ir} A_{rj}$ of them.

Every walk is counted for exactly one r, so the number of walks of length $k+1$ from n_i to n_j is $\sum_r (A^k)_{ir} A_{rj}$, which is $(A^{k+1})_{ij}$ by eq. (3). □

Why the proof is valid, and why it counts walks: it uses only that a walk is a vertex sequence whose consecutive vertices are adjacent, repeats allowed. Sorting by the last-but-one vertex therefore never has to ask whether n_r was visited before. For paths the deletion of the final vertex would still give a path, but appending n_j may repeat a vertex, and the one to one correspondence breaks; example 4.5 shows a walk that is counted and is not a path. The same induction works for directed graphs and multigraphs [2, Proposition 2, p. 1], which are outside this note's scope.

4.2 The board graph, worked

In plain words. Take three islands in a row, with a bridge between each neighbouring pair. Multiply the table by itself once, twice, and more times. Each cell is a number of trips. We list the trips behind some cells to check that the numbers are right.

Example 4.2 (Powers of the board matrix)**.** For the board graph (fig. 1, left) the powers, computed by script (appendix A), are

$$A^2 = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{pmatrix}, \quad A^3 = \begin{pmatrix} 0 & 2 & 0 \\ 2 & 0 & 2 \\ 0 & 2 & 0 \end{pmatrix}, \quad A^4 = \begin{pmatrix} 2 & 0 & 2 \\ 0 & 4 & 0 \\ 2 & 0 & 2 \end{pmatrix}.$$

Each entry counts walks, and the script listed them:

- $(A^2)_{11} = 1$: the walk $n_1 n_2 n_1$.
- $(A^2)_{13} = 1$: the walk $n_1 n_2 n_3$.
- $(A^2)_{22} = 2$: the walks $n_2 n_1 n_2$ and $n_2 n_3 n_2$.
- $(A^3)_{12} = 2$: the walks $n_1 n_2 n_1 n_2$ and $n_1 n_2 n_3 n_2$.
- $(A^4)_{22} = 4$: the walks $n_2 n_1 n_2 n_1 n_2$, $n_2 n_1 n_2 n_3 n_2$, $n_2 n_3 n_2 n_1 n_2$, and $n_2 n_3 n_2 n_3 n_2$.
- $(A^2)_{12} = 0$ and $(A^3)_{11} = 0$: there is no such walk.

The inductive case for $(A^3)_{12}$ reads, with $r = 1,2,3$, $\sum_r (A^2)_{1r} A_{r2} = (A^2)_{11} A_{12} + (A^2)_{12} A_{22} + (A^2)_{13} A_{32} = 1 \cdot 1 + 0 \cdot 0 + 1 \cdot 1 = 2$: the term for $r = 1$ is the walk $n_1 n_2 n_1$ followed by the move to n_2, and the term for $r = 3$ is $n_1 n_2 n_3$ followed by the move to n_2.

The zero entries have a reason, which is a fact about this graph only: every edge joins n_2 to n_1 or n_3, so each hop alternates between the middle vertex n_2 and an end vertex. A walk from n_1 to n_1 makes an even number of hops, and a walk from n_1 to n_2 an odd number. As a check of instances and not as a proof, the script also compared the matrix power with a direct enumeration of walks for $k = 0, \ldots, 6$ on six small graphs and for $k = 0, \ldots, 5$ on all 64 labelled simple graphs on four vertices ($64 = 2^6$ choices of which of the 6 possible pairs are edges), and found no mismatch.

Non-example 4.3 (What the theorem does not say)**.** The entry $(A^k)_{ij}$ is not "1 if n_j can be reached from n_i". In the board graph $(A^4)_{22} = 4$, not 1.

4.3 Properties of the powers

In plain words. A trip can be run backwards. So the table for any number of hops is the same when flipped. For two hops, the cell for an island and itself is the number of bridges at that island. Each bridge gives one trip out and back.

Proposition 4.4 (Symmetry and the diagonal of A^2)**.** *For every $k \geq 0$ the matrix A^k is symmetric. Moreover $(A^2)_{ii} = \deg(n_i)$ for every i, and so the sum of the diagonal entries of A^2 is $2|E|$.*

The second statement is made, as an equality of walk counts, in the University of Galway notes [7, p. 4], whose sentence reads "The number of walks of length 2 from a vertex to itself is just the number of edges at that vertex, or the vertex degree."; the proof here is the note's own.

Proof. Since $A^T = A$ and $(XY)^T = Y^T X^T$, we get $(A^k)^T = (A^T)^k = A^k$. In words: reversing a walk from n_i to n_j gives a walk from n_j to n_i of the same length, and this is one to one. For the second statement, theorem 4.1 says $(A^2)_{ii}$ counts the walks $n_i n_r n_i$. Such a walk exists for each vertex n_r adjacent to n_i and for no other, so the count is $\deg(n_i)$. The diagonal sum is the sum of all degrees, which is $2|E|$ by proposition 2.17. □

4.4 Walks are not paths

In plain words. The table counts trips. It does not count fresh-island trips. Take the frog that hops from the first island to the middle one and back. That is a trip of two hops from the first island to itself. The frog lands on the first island twice, so it is not a fresh-island trip. The table counts it anyway.

Example 4.5 (The entry $(A^2)_{11}$ in the board graph)**.** By example 4.2, $(A^2)_{11} = 1$, and the walk it counts is $n_1 n_2 n_1$. This walk visits n_1 twice, so it is not a path (definition 3.13). No path of length 2 joins n_1 to itself: by the remark after definition 3.17 the only path from n_1 to n_1 is (n_1), of length 0. So "$(A^2)_{11}$ counts the paths of length 2 from n_1 to n_1" is false, while "counts the walks" is true. Likewise $(A^3)_{12} = 2$ counts $n_1 n_2 n_1 n_2$ and $n_1 n_2 n_3 n_2$, but a path of length 3 needs four distinct vertices and the graph has three.

Where walks and paths do agree: $(A^2)_{13} = 1$ counts $n_1 n_2 n_3$, which is a path. In general, if k equals the distance between n_i and n_j, every walk of length k from n_i to n_j is a path, since a repeated vertex would let us cut out a closed block and get a shorter walk (lemma 3.30), contradicting that k is the least length; so then $(A^k)_{ij}$ is the number of paths of length k from n_i to n_j. For k larger than the distance the entry is at least the number of paths of length k and can be larger, as in example 4.5, where there are walks and no paths at all; it can also agree, as in the triangle, where $(A^2)_{12} = 1$ counts the single walk $n_1 n_3 n_2$, which is a path.

Remark 4.6 (What a positive entry gives). If $(A^k)_{ij} > 0$, there is a walk of length k from n_i to n_j (theorem 4.1), hence, by lemma 3.30, a path from n_i to n_j of length at most k. It does not follow that a path of length exactly k exists, and a path of exactly that length can be missing. This is what section 5 uses.

4.5 Distance from the powers

In plain words. Pick two different islands. To see how far apart they are, look at the tables for one hop, two hops, and so on. Stop at the first table in which the cell for the two islands is not zero. That number of hops is the fewest hops between them.

Corollary 4.7 (Distance from walk counts)**.** *For vertices $n_i \neq n_j$, the distance $d(n_i, n_j)$ is the least $k \geq 1$ with $(A^k)_{ij} > 0$ if such a k exists, and ∞ otherwise. Also $d(n_i, n_i) = 0$.*

Paramadevan and Sotheeswaran state this for connected graphs [9, Theorem 3, p. 17]; the proof below is the note's own (see section 6 for what their proof reads into a non-zero entry).

Proof. By theorem 4.1, $(A^k)_{ij} > 0$ exactly when a walk of length k from n_i to n_j exists. So the least such k is the least length of a walk from n_i to n_j, which equals $d(n_i, n_j)$ by proposition 3.34; if no such k exists no walk exists, and both sides are ∞. For $i = j$ the walk (n_i) has length 0, so $d(n_i, n_i) = 0$. □

Example 4.8. In the board graph, $(A^1)_{13} = 0$ and $(A^2)_{13} = 1$, so $d(n_1, n_3) = 2$, as in definition 3.25.

5 Question 2: deciding connectivity from A

Question (Q2) asks how to know from A whether the graph is connected. This section defines connectedness and components, proves a test that uses only powers of A, and shows it on one connected and one disconnected graph.

5.1 Connected graphs and components

In plain words. A pond is in one piece when the frog can hop from any island to any other island. It may cross many bridges on the way. If some island cannot be reached from some other island, the pond is not in one piece.

Definition 5.1 (Connected; textbook Definition 2.1.1, and the board)**.** G is *connected* if for any two vertices u and v there is a path from u to v. The board states the same definition: for any n_i, n_j there exists a path between the nodes [5, 6]. The textbook's Definition 2.1.1 asks for a (u, v)-"path" [10, Definition 2.1.1, p. 48]. (Asking for a walk, a trail, or a path gives the same connected graphs; see proposition 3.33.)

Remark 5.2. Two readings of "path" are in play. The textbook's "path" is a trail (definition 3.9); the course's, and this note's, is the path of definition 3.13 (Convention 5). By proposition 3.33 both readings, and a reading by walks, give the same connected graphs, so the choice does not affect this section. A graph with one vertex is connected, since the path (u) of length 0 joins u to itself.

Example 5.3. The board graph is connected: n_1n_2, n_2n_3, and $n_1n_2n_3$ are paths that join its three pairs of different vertices. The textbook's three graphs under its Fig. 2.1.1 are connected [10, p. 48].

Non-example 5.4. The graph with the single edge n_1n_2 and the isolated vertex n_3 is not connected: no walk, hence no path, joins n_3 to n_1. The graph of the textbook's Fig. 2.1.2 is not connected and has four connectivity components [10, p. 49].

In plain words. Call two islands friends if a frog can get from one to the other. Every island is its own friend. If the first island is a friend of the second, the second is a friend of the first. A friend of a friend is a friend.

A *relation* $\sim$ on the vertices says, for each ordered pair u, v, whether $u \sim v$ holds or not. It is *reflexive* if $u \sim u$ for every u, *symmetric* if $u \sim v$ implies $v \sim u$, and *transitive* if $u \sim v$ and $v \sim w$ imply $u \sim w$. A relation with all three properties is an *equivalence relation*; it splits the set into *equivalence classes*, the groups of vertices that are related to each other, so that every vertex lies in exactly one class.

Lemma 5.5 (Being joined by a path is an equivalence relation)**.** *Let $u \sim v$ mean that there is a path from u to v. Then $\sim$ is reflexive, symmetric, and transitive.*

The textbook notes these three properties on printed p. 49 without giving a proof [10]; the proof below is the note's own.

Proof. Reflexive. The walk (u) of length 0 is a path from u to u. *Symmetric.* Reversing a path from u to v gives a path from v to u. *Transitive.* If W_1 is a walk from u to v and W_2 is a walk from v to w, writing W_1 and then W_2 (with the repeated v written once) is a walk from u to w, and by lemma 3.30 it contains a path from u to w. □

In plain words. The islands split into groups of friends. Nobody in one group is a friend of anybody in another group.

Definition 5.6 (Connectivity component; textbook, printed p. 49)**.** The equivalence classes of $\sim$, each taken together with the edges of G that join two of its vertices (the subgraph *induced* on the class), are the *connectivity components* of G. The textbook introduces them in the prose after Definition 2.1.1, with no number of their own [10, p. 49]. Two vertices lie in the same component exactly when a path joins them.

Remark 5.7. Every vertex lies in exactly one component, and a graph with $N \geq 1$ vertices has at least one component.

Example 5.8. The board graph has one component, $\{n_1, n_2, n_3\}$.

Non-example 5.9. The graph with the single edge n_1n_2 and the isolated vertex n_3 has two components, $\{n_1, n_2\}$ and $\{n_3\}$. The set $\{n_1\}$ alone is not a component of it, because n_2 is joined to n_1 by a path.

In plain words. The pond is in one piece exactly when all the islands form a single group of friends.

Corollary 5.10 (Textbook Remark 2.1.1)**.** *G is connected if and only if it has exactly one connectivity component.*

The textbook states this as Remark 2.1.1 [10, p. 49]; the argument here is the note's own.

Proof. G is connected exactly when every two vertices are related by $\sim$, that is, exactly when there is only one equivalence class. Since $N \geq 1$, at least one class exists. □

Components are a reformulation of connectedness (corollary 5.10). The proofs of the two tests below do not use them; they are used only to describe the shape of the matrices in example 5.15.

5.2 The test

In plain words. Pick two different islands. Make tables for one hop, for two hops, and so on. Stop at one fewer hops than there are islands. Look at the cell for the two islands in each table. The frog can get from one island to the other exactly when one of these cells is not zero. A longer list is never needed.

Lemma 5.11 (Reachability from the powers)**.** *Let $i \neq j$. There is a path from n_i to n_j if and only if $(A^k)_{ij} > 0$ for some k with $1 \leq k \leq N-1$.*

The reasoning combines walk counting with the walk-to-path lemma; it is the note's own proof, and the step from a positive entry to a path is the one that some sources pass over (section 6).

Proof. ($\Rightarrow$) Let a path from n_i to n_j have length k. Since $i \neq j$, $k \geq 1$. By corollary 3.32, $k \leq N-1$. A path is a walk, so by theorem 4.1 the entry $(A^k)_{ij}$ counts at least this walk, and $(A^k)_{ij} \geq 1$. ($\Leftarrow$) If $(A^k)_{ij} > 0$ with $1 \leq k \leq N-1$, then by theorem 4.1 some walk of length k joins n_i to n_j. By lemma 3.30 it contains a path from n_i to n_j, of length at most $k \leq N-1$. □

In plain words. Add up the tables for one hop, for two hops, and so on. Stop at one fewer hops than there are islands. Look only at cells where the row island and the column island are different. The other cells count trips that come back to where they started, and those say nothing about reaching other islands. If none of the cells you look at is zero, the pond is in one piece. If one of them is zero, it is not.

Theorem 5.12 (Sum test)**.** *Let $B = A + A^2 + \cdots + A^{N-1}$. Then G is connected if and only if $B_{ij} > 0$ for all $i \neq j$.*

The statement is Theorem 2 of Paramadevan and Sotheeswaran [9, p. 17]; the proof here is the note's own.

Proof. Each A^k has non-negative entries (remark 2.19), so $B_{ij} = \sum_{k=1}^{N-1} (A^k)_{ij}$ is a sum of non-negative terms, no term can cancel another, and the sum is positive exactly when at least one term is positive. For $i \neq j$ this holds exactly when a path joins n_i and n_j, by lemma 5.11. So $B_{ij} > 0$ for all $i \neq j$ exactly when every pair of different vertices is joined by a path, which is definition 5.1 (a vertex is joined to itself by the path of length 0). For $N = 1$ the sum is empty and there are no entries off the diagonal; G is connected and the statement holds without content. □

In plain words. There is a shortcut that needs only one new table. Start with the island table. Add a one to every cell where the row island and the column island are the same. Multiply this new table by itself again and again. Use it as a factor one fewer time than there are islands. The pond is in one piece exactly when no cell of the result is zero.

Theorem 5.13 (Power test)**.** *G is connected if and only if every entry of $T = (I + A)^{N-1}$ is positive.*

The statement is Proposition 3 of Gotti [2, p. 2] and Theorem 2 of Bai [1, p. 1]. Gotti's proof uses the same binomial identity as the proof below [2, p. 2]. That this test is equivalent to the sum test is shown here, and no source read states both together.

Proof. Two matrices X and Y *commute* if $XY = YX$. The binomial theorem $(X + Y)^n = \sum_k \binom{n}{k} X^k Y^{n-k}$ holds for matrices that commute, and I commutes with every matrix, in particular with A (because $IA = A = AI$). The binomial coefficient $\binom{n}{k}$ is the number of ways to choose k items from n; it is a positive integer for $0 \leq k \leq n$. So

$$(I + A)^{N-1} = \sum_{k=0}^{N-1} \binom{N-1}{k} A^k, \tag{4}$$

where each binomial coefficient is at least 1. Take $i \neq j$. Then $I_{ij} = 0$, so $T_{ij} = \sum_{k=1}^{N-1} \binom{N-1}{k} (A^k)_{ij}$. All terms are non-negative and all coefficients are at least 1, so $T_{ij} > 0$ exactly when some $(A^k)_{ij}$ with $1 \leq k \leq N-1$ is positive, that is, exactly when $B_{ij} > 0$. On the diagonal the term $k = 0$ contributes $\binom{N-1}{0} I_{ii} = 1$, so every diagonal entry of T is at least 1 whatever the graph. Hence "no entry of T is zero" means "$B_{ij} > 0$ for all $i \neq j$", and theorem 5.12 finishes the proof. □

So the answer to (Q2) is: form B, or T; look at the entries off the diagonal of B, or at all entries of T; the graph is connected if and only if none of them is zero. The entries are walk counts and not path counts; only whether they are zero is used.

5.3 What the test does not say

- The diagonal of B is ignored. A connected graph can have zeros on the diagonal of B: for $N = 2$ and one edge, $B = A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. The reason: B_{ii} counts closed walks from n_i back to itself, which say nothing about reaching other vertices; with $N = 2$ the sum stops at A^1, which has no closed walks. The disconnected example below has $B_{33} = 0$ as well, but what makes it disconnected are the zeros off the diagonal.

- A positive entry says a walk of exactly length k exists, hence a path of length at most k (remark 4.6); it does not say a path of exactly length k exists.
- The sum has to go up to A^{N-1}. Stopping earlier can give a false "not connected" (example 5.17).

5.4 Worked examples

In plain words. We run the test on two ponds with three islands. In the first, the islands lie in a row, and the frog can get anywhere. In the second, two islands share a bridge and the third island stands alone. The tables are computed by a program, not by hand.

Example 5.14 (A connected graph: the board graph)**.** Here $N = 3$, so $B = A + A^2$ and $T = (I + A)^2$. With A and A^2 from example 4.2,

$$B = A + A^2 = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 1 \end{pmatrix}, \qquad T = (I + A)^2 = \begin{pmatrix} 2 & 2 & 1 \\ 2 & 3 & 2 \\ 1 & 2 & 2 \end{pmatrix}.$$

All entries of B off the diagonal are positive: the pairs n_1n_2 and n_2n_3 get theirs from A, and the pair n_1n_3 gets $B_{13} = 1$ from A^2, which counts the path $n_1n_2n_3$. So G is connected. T has no zero entry; each entry equals the entry of $A^0 + 2A + A^2$, for example $T_{12} = 0 + 2 + 0 = 2$ and $T_{22} = 1 + 0 + 2 = 3$, as eq. (4) says. The number $B_{22} = 2$ counts the walks $n_2n_1n_2$ and $n_2n_3n_2$ and not paths; only its being positive matters, and the diagonal is not read.

Example 5.15 (A disconnected graph)**.** Take the edge n_1n_2 and the isolated vertex n_3 (fig. 1, right). Then

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad A^2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad T = \begin{pmatrix} 2 & 2 & 0 \\ 2 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

The entries $B_{13} = B_{23} = B_{31} = B_{32} = 0$, so G is not connected by theorem 5.12. T has zeros in the same four places, so theorem 5.13 says the same. The two components of corollary 5.10 are $\{n_1, n_2\}$ and $\{n_3\}$, and the non-zero entries of B and T all lie in the rows and columns of one of them: they form two blocks along the diagonal, a 2×2 block and a 1×1 block. In general no walk joins vertices in different components, so by lemma 5.11 every entry that links two components is zero.

In plain words. Suppose we looked only at the table for two hops, with three islands in a row. The first island and the middle island are next door. Yet their cell is zero. Every trip between them has an odd number of hops.

Example 5.16 (A single power is not a test)**.** In the board graph the only power tried by a test that used A^{N-1} alone would be A^2, and $(A^2)_{12} = 0$ although n_1n_2 is an edge. The reason in words: every edge of this graph joins n_2 to n_1 or n_3, so every walk from n_1 to n_2 has odd length, and A^2 sees only even lengths.

In plain words. Now take four islands in a row. The first and the last island are three hops apart. A list of tables that stops at two hops misses them. So the list must go on until one fewer hops than there are islands.

Example 5.17 (Stopping the sum early)**.** For the path graph with edges n_1n_2, n_2n_3, n_3n_4 ($N = 4$), the script gives $(A^1)_{14} = 0$, $(A^2)_{14} = 0$, and $(A^3)_{14} = 1$. The entry $(1,4)$ of $A + A^2$ is 0, and of $A + A^2 + A^3$ it is 1. A sum stopped at A^2 would call this connected graph disconnected.

5.5 Breadth-first search, for comparison

In plain words. There is another way to decide, with no tables multiplied. Stand on one island. Mark it. Then mark every island next door to a marked island, round after round, until no new island is marked. The pond is in one piece exactly when every island ends up marked.

Breadth-first search (BFS) from a start vertex s visits s, then all of its neighbours, then all not yet visited neighbours of those, layer by layer, and stops when a layer is empty. Lau's lecture slides list "To determine whether a given graph is connected" among the questions BFS answers, and state that there is a path from s to v if and only if v is marked visited at the end [4, pp. 4, 8]; McCauley's slides state that in a connected undirected graph BFS from any vertex visits every vertex, and that if the graph is not connected BFS will not visit all nodes [8, pp. 10, 7]. Together this suggests the test: run BFS from n_1; G is connected exactly when all N vertices are visited. This is the note's interpretation of the two sources for its own definition of connected, which neither source states in exactly this form; the BFS lemma is cited and not proved here. BFS reads A row by row and needs no matrix powers. As an estimate by counting, not a measurement: BFS reads one row of N entries for each of at most N visited vertices, about N^2 operations. A product of two $N \times N$ matrices has N^2 entries, each a sum of N products by eq. (2), so about N^3 multiplications, and reaching A^{N-1} from A takes $N-2$ products. So the matrix tests answer the question as posed, with A alone, but are not the cheap way.

6 Discussion

6.1 Checks by computation

The proofs above do not depend on the following checks; they are checks of instances. A script compared the sum test, the power test, and a breadth-first search written independently of the matrices on every labelled simple graph with up to six vertices ("labelled" means that the vertices have fixed names, so each choice of which pairs are edges is a separate graph, $2^{N(N-1)/2}$ in all), and the three tests agreed on every one. A second script checked the cut of lemma 3.30 on all walks up to a fixed length on all graphs with three, four, and five vertices, with no failure; it also found that no path was longer than $N-1$ and that shortest walks, trails, and paths gave the same distance. The counts are in appendix A.

6.2 The board drawing and the board matrix

The board drawing and the board matrix differ edge by edge. The drawing shows three lines: between 1 and 2 (with an arrowhead at 2), between 1 and 3, and between 2 and 3. The matrix has ones for the pairs n_1n_2 and n_2n_3 only and $A_{13} = A_{31} = 0$, so the line between 1 and 3 is in the drawing and not in the matrix [5, 6]. The note uses the matrix (Convention 4) and reads the arrowhead as a stray mark, because the matrix is symmetric and the graph is undirected. Read literally, the drawing is the triangle on n_1, n_2, n_3. The script gives, for the triangle,

$$A^2 = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}, \quad T = (I+A)^2 = \begin{pmatrix} 3 & 3 & 3 \\ 3 & 3 & 3 \\ 3 & 3 & 3 \end{pmatrix}.$$

Both tests again say that the graph is connected, so the answer to (Q2) for the board does not depend on which of the two graphs is meant. The entries differ, however: $(A^2)_{11} = 2 = \deg(n_1)$ here, against 1 for the board graph, in agreement with proposition 4.4.

6.3 Words that differ between sources

The same word names different things in the textbook and in English (table 2), and a reader who moves between them meets three traps.

- The textbook's "path" is a trail. Its Definition 2.1.1 of connected and its Lemma 1.2.1 use it [10, pp. 23, 48]. Proposition 3.33 shows that this does not change which graphs are connected.
- The textbook's "closed path or cycle" is a closed trail, which is wider than a cycle: two triangles sharing a vertex give a closed trail of length 6 that is not a cycle (the remark after definition 3.21). This note uses "cycle" only in the narrow sense.
- The textbook measures distance by shortest trails; proposition 3.34 shows that this equals the shortest walk and the shortest path.

6.4 Statements in two sources that need care

This subsection is optional. It explains why some sentences in published sources can mislead a reader who takes "path" to mean a walk with no repeated vertex. The walk-counting theorem and the connectivity tests are stated correctly in the sources named at theorems 4.1, 5.12 and 5.13, read with walks. Three sentences are quoted below, each restated in this note's notation (the superscripts in the first quotation are as the text layer of the source renders them). Nothing else in the note depends on them.

- Paramadevan and Sotheeswaran, Theorem 1: "the number of different paths of length l between the ith and j th vertices of G is equal to the (i, j)th entry of AlG" [9, p. 16]. In this note's notation: $(A^l)_{ij}$ counts the paths of length l from n_i to n_j. The paper never defines "path". Read as walks this is theorem 4.1. Read as paths in the sense of definition 3.13 it is false: in the board graph $(A^2)_{11} = 1$, but no path of length 2 joins n_1 to itself (example 4.5).
- The proof of their Theorem 2 says "bij = number of different paths of length less than n from the vertex i to j" [9, p. 17]. In this note's notation: B_{ij} counts the paths of length less than N from n_i to n_j. Read as paths in the sense of definition 3.13, it fails for the board graph: $B_{22} = 2$ (example 5.14), but at most one path of any length joins n_2 to itself, the one-vertex path (n_2) (and none if length 0 is not allowed). The statement of Theorem 2 itself is the sum test and is correct (theorem 5.12); what the proof lacks is the step from a non-zero entry to a path, which is lemma 3.30.
- Bai's proof sketch of Theorem 2 says that "$(I+A)^{n-1}$ aggregates all paths of length $\leq n-1$" [1, p. 2]. In this note's notation: T_{ij} counts the paths of length at most $N-1$ from n_i to n_j. Bai defines a path as "a walk where no vertex is repeated (except possibly the first and last vertex if it's a cycle)" [1, p. 1]. For the board graph $T_{11} = 2$ (example 5.14), but at most one path of any length joins n_1 to itself, the one-vertex path (n_1); the walk $n_1 n_2 n_1$ is not a cycle in the sense of definition 3.21, since it reuses an edge. Bai's own convention for a cycle was not read, so this refutation is made for the restatement in this note's notation. Bai's Theorem 1 states that the entry counts "walks — not paths" [1, p. 1], and Bai's statement of Theorem 2 is correct, as theorem 5.13 shows.

6.5 Limits and open points

- The mark above A on the board stays a reading (Convention 3); only the author of the board can settle it. The theorem is stated for all k.
- The report of the assignment, including the course's use of "path", is a summary and not the assignment's own words [11]. The textbook definition itself is firm [10, Definition 1.2.13, p. 22].
- The textbook was searched, not read in full, for a walk-counting theorem and a matrix test (Convention 7).
- The English column of table 2 rests on two sources, and whether other English books use the same words was not checked. The word "trail" for the textbook's "path" is this note's choice.

- For breadth-first search, Lau's slides do not define "path" on the pages read and their proof of the lemma is on image slides that were not read, and McCauley's converse ("not connected, so not all nodes visited") is a slide remark and not a stated lemma [4, 8]. The BFS test is therefore stated as a comparison and as an interpretation of the two sources. A standard algorithms textbook was not opened and would settle it. The cost remark is an estimate by counting.
- The note does not treat directed graphs, multigraphs, weights, or eigenvalue methods, which are outside its scope.

7 Conclusion

The three questions have direct answers.

(Q1) For a finite simple undirected graph with adjacency matrix A and every $k \geq 0$, the entry $(A^k)_{ij}$ is the number of walks of length k from n_i to n_j (theorem 4.1). It counts walks, in which vertices and edges may repeat, and not paths: in the board graph $(A^2)_{11} = 1$ counts $n_1 n_2 n_1$, which is not a path (example 4.5). It is not "1 if reachable" either (non-example 4.3). A positive entry does give a path of length at most k (remark 4.6).

(Q2) G is connected if and only if every entry off the diagonal of $B = A + A^2 + \cdots + A^{N-1}$ is positive (theorem 5.12), and also if and only if $(I + A)^{N-1}$ has no zero entry (theorem 5.13). The cutoff $N - 1$ is the largest length a path can have: a path visits each vertex at most once, so it has at most $N - 1$ edges. The proof uses walk counting, the fact that a walk contains a path no longer than itself (lemma 3.30), and the fact that a path has length at most $N - 1$ (corollary 3.32). The board graph, whose edges are $n_1 n_2$ and $n_2 n_3$, is connected (example 5.14); one edge and an isolated vertex is not (example 5.15). A single power is not a test (example 5.16).

(Q3) A path between n_i and n_j is a sequence $n_i = u_0, u_1, \ldots, u_k = n_j$ with $k \geq 0$ in which u_{t-1} and u_t are adjacent for every t and the vertices $u_0, \ldots, u_k$ are pairwise distinct; this is the textbook's simple path, Definition 1.2.13 (definition 3.17). The textbook's own word "path" means a trail. The two readings define the same connected graphs (proposition 3.33).

References

[1] Shuliang Bai. *Important Properties and Applications of the Adjacency Matrix (20250310)*. BIMSA lecture note, dated 2025-03-10. 2025. URL: `https://bimsa.net/doc/notes/48535.pdf` (visited on Nov. 6, 2026).

[2] Felix Gotti. *MIT 18.211 Combinatorial Analysis, Lecture 27: Adjacency Matrices and The Matrix-Tree Theorem*. MIT course lecture notes. No usable date in the file. URL: `https://math.mit.edu/~fgotti/docs/Courses/C.%20Combinatorial%20Analysis/27.%20The%20Matrix-Tree%20Theorem/The%20Matrix-Tree%20Theorem.pdf` (visited on Nov. 6, 2026).

[3] Darij Grinberg. *An introduction to graph theory (Text for Math 530, Drexel University)*. arXiv:2308.04512v3 [math.HO], version of June 7, 2025. 2025. URL: `https://arxiv.org/abs/2308.04512` (visited on Nov. 6, 2026).

[4] Lap Chi Lau. *CS 341 - Algorithms, Lecture 5 - Breadth First Search (lecture slides, 26 May 2021)*. University of Waterloo, CS 341 lecture slides. 2021. URL: `https://cs.uwaterloo.ca/~lapchi/cs341-2021/notes/L05s.pdf` (visited on Nov. 6, 2026).

[5] *Lecture whiteboard photo 1*. Local file `sources/whiteboard-1.jpg`. Photo supplied with the request for this note; handwritten board, viewed as an image.

[6] *Lecture whiteboard photo 2*. Local file `sources/whiteboard-2.jpg`. Photo supplied with the request for this note; the same board as photo 1 at a different angle.

[7] *Linear algebra notes, 1.1 The Adjacency Matrix (University of Galway notes)*. University of Galway, Mathematics, lecture notes, chapter 1 section 1.1. The file carries no author name and no date. URL: https://maths.nuigalway.ie/~rquinlan/linearalgebra/section1-1.pdf (visited on Nov. 6, 2026).

[8] Sam McCauley. *Lecture 4: BFS, Graph Representations, DFS (lecture slides, February 17, 2025)*. Williams College, algorithms course lecture slides. 2025. URL: https://williams-cs.github.io/cs256-s25/lectures/04-graphs.pdf (visited on Nov. 6, 2026).

[9] P. Paramadevan and S. Sotheeswaran. "Properties of Adjacency Matrix of a Graph and It's Construction". In: *JSc EUSL* 12.1 (2021). Eastern University, Sri Lanka, pp. 13–21. URL: https://www.fsc.esn.ac.lk/jsc/archive/12.1.2.pdf (visited on Nov. 6, 2026).

[10] P. A. Petrosyan, V. V. Mkrtchyan, and R. R. Kamalyan. *Graph Theory, study guide*. In Armenian; transliterated title Grafneri tesutyun. Local file Grafneri_tesutyun.pdf; Yerevan State University Press. Course textbook, supplied with the request for this note. Printed page numbers are one lower than PDF page numbers. 2015.

[11] *Summary of user statements*. Local file sources/user-statements.md. Summary recorded on 2026-11-06, supplied with the request for this note; a summary, not the assignment's own words. 2026.

A Sources, pages checked, and reproduction

This appendix is the only place that describes how the note was prepared. The mathematical sections only cite.

What was read.

- The two whiteboard photos were viewed as images and read together; they show one board [5, 6]. Quotations of the board are a transcription of handwriting.
- The textbook, a PDF of 216 pages in Armenian [10]. Printed pages are one lower than PDF pages; this note cites printed pages. Printed pages 7, 9, 11, 16, 21, 22, 23, 48, and 49 were viewed as images. The text layer drops mathematical symbols, so formulas were checked on the images. Other pages were reached only through a text search for the Armenian phrase for "adjacency matrix", which finds it on printed pages 7, 11, 12, 34, and 61; pages 7 and 11 were read and hold the notation and the definition, and pages 12, 34, and 61 were not read for this note. No walk-counting theorem and no matrix test for connectivity was found, which is the result of a search and not of reading the whole book (Convention 7).
- University of Galway notes, section 1.1, read in full; they carry no author name and no date [7]. Gotti's lecture notes, read in full; no usable date [2].
- Grinberg, an arXiv text of 454 pages; Section 2.9 and the breadth-first-search passage near p. 176 were read, and Definition 2.10.1, Exercise 2.16(d), and Definition 3.4.5 from the text layer [3]. Page numbers are the version's own.
- Bai's lecture note, Section 1 only [1]. Paramadevan and Sotheeswaran, Sections 2.2 and 3.1 [9].
- Lau's slides, read through their text layer, with pages 3, 4, and 8 used here; the algorithm listing and the correctness proof are images and were not read [4]. McCauley's slides, text layer, pages 5 to 12 and 24 [8]. Page numbers of the two slide decks are PDF pages.
- The summary of the assignment [11] is a summary written when the request was received; it is not quoted as the assignment's words.

Counts behind the checks of section 6. The script `scripts/m3_connectivity.py` compared the sum test, the power test, and a breadth-first search on all labelled simple graphs with $N = 1, \ldots, 6$ vertices, that is 1, 2, 8, 64, 1024, and 32768 graphs, of which 1, 1, 4, 38, 728, and 26704 are connected, with no disagreement. The counts of connected graphs are what the script found, not figures taken from a source. The script `scripts/walk_path_lemmas.py` checked the cut of lemma 3.30 on every walk of length 0 to 8 on all 8 graphs with three vertices, of length 0 to 7 on all 64 graphs with four, and of length 0 to 5 on all 1024 graphs with five (2052, 72664, and 743680 walks), with no failure.

How to rerun the computations. From the paper folder:

- `python3 -I scripts/walks_check.py` prints the powers of the board matrix, the walks behind the entries, the classification of every example sequence on H, the path graph on four vertices, and the brute force check of theorem 4.1; its output is in `scripts/output-walks_check.txt`.
- `python3 -I scripts/m3_connectivity.py` prints A, A^2, B, and T for the connected, the disconnected, and the triangle example, the check that a single power is not a test, the binomial identity, and the exhaustive comparison of the three tests for $N = 1, \ldots, 6$; its output is in `scripts/output-m3_connectivity.txt`.
- `python3 -I scripts/walk_path_lemmas.py` checks lemmas 3.29 and 3.30, corollary 3.32, and proposition 3.34 on all graphs with three, four, and five vertices (about six seconds); its output is in `scripts/output-walk_path_lemmas.txt`.

All three use only the Python standard library. The note itself compiles with `latexmk` and `biber` (TeX Live) by the commands in the paper folder's `README.md`.